WORKED SOLUTION

STEP II 2007, Question 12

Probability and Statistics • Geometric Distribution Maximum

STEP II 2007, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q12

The key idea

Forget the rather fussy description of the procedure and look at each die on its own. Each die is thrown until it first shows a six, and the experiment ends when the *slower* of the two dice has got there. So the number of throws is the maximum of two independent geometric random variables, and the maximum is easiest to handle through the cumulative probability $\mathrm{P}(\max \le r)$.

Solution

Part (i)

Let X be the number of the throw on which the first die first shows a six, and define Y in the same way for the second die. Every throw of a die is independent of every other throw, so X and Y are independent, and each is geometric:

$$\mathrm{P}(X = k) = q^{k-1}p, \qquad \mathrm{P}(X \le r) = 1 - \mathrm{P}(\text{first } r \text{ throws all non-six}) = 1 - q^r.$$

How does the procedure fit this? While neither die has shown a six we throw both; once one has, we keep throwing only the other one. Either way, each die is thrown exactly until its first six (and never after it), and we stop as soon as both have shown a six. If they first show a six on the same throw, that is the case where both show a six together. So the number of throws is

$$N = \max(X, Y).$$

? **Ask yourself** Why work with $\mathrm{P}(N \le r)$ rather than $\mathrm{P}(N = r)$ directly?

The event $N \le r$ is simply "$X \le r$ *and* $Y \le r$", and independence turns that into a product. The event $N = r$ splits into awkward cases (one die finishes at r and the other earlier, or both at r). So

$$\mathrm{P}(N \le r) = \mathrm{P}(X \le r)\,\mathrm{P}(Y \le r) = (1 - q^r)^2,$$

and, for $r \geq 1$,

$$\begin{aligned}
\mathrm{P}(N = r) &= \mathrm{P}(N \leq r) - \mathrm{P}(N \leq r-1) \\
&= (1-q^r)^2 - (1-q^{r-1})^2 \\
&= \left[(1-q^r) - (1-q^{r-1})\right]\left[(1-q^r) + (1-q^{r-1})\right] \\
&= (q^{r-1} - q^r)(2 - q^{r-1} - q^r) \\
&= pq^{r-1}\,(2 - q^{r-1} - q^r),
\end{aligned}$$

using the difference of two squares and then $q^{r-1} - q^r = q^{r-1}(1-q) = pq^{r-1}$. This is the required result.

For the expectation, split the probability into its two pieces:

$$\mathrm{P}(N = r) = 2pq^{r-1} - pq^{2r-2}(1+q).$$

Differentiating the given result (or dividing it by x) gives $\sum_{r\geq 1} rx^{r-1} = (1-x)^{-2}$ for $|x| < 1$. Using it with $x = q$ and with $x = q^2$ (both less than 1 for $0 < p \leq 1$):

$$\begin{aligned}
\mathrm{E}(N) &= 2p\sum_{r=1}^{\infty} rq^{r-1} - p(1+q)\sum_{r=1}^{\infty} r(q^2)^{r-1} \\
&= \frac{2p}{(1-q)^2} - \frac{p(1+q)}{(1-q^2)^2} \\
&= \frac{2}{p} - \frac{p(1+q)}{p^2(1+q)^2} \\
&= \frac{2}{p} - \frac{1}{p(1+q)} \\
&= \frac{1+2q}{p(1+q)},
\end{aligned}$$

where we used $1 - q = p$ and $1 - q^2 = (1-q)(1+q) = p(1+q)$. In terms of p alone (put $q = 1-p$),

$$\mathrm{E}(N) = \frac{3-2p}{p(2-p)}.$$

Part (ii)

Over many experiments the mean number of throws should be close to $\mathrm{E}(N)$, so we estimate p by solving $\mathrm{E}(N) = m$:

$$m\,p(2-p) = 3 - 2p \quad \Longleftrightarrow \quad mp^2 - 2(m+1)p + 3 = 0,$$

so

$$p = \frac{(m+1) \pm \sqrt{(m+1)^2 - 3m}}{m} = \frac{m+1 \pm \sqrt{m^2 - m + 1}}{m}.$$

Ask yourself Which root do we want?

Every experiment takes at least one throw, so $m \geq 1$. Then $\sqrt{m^2 - m + 1} \geq 1$ (because $m^2 - m = m(m-1) \geq 0$), so the root with the plus sign is at least $(m+2)/m > 1$, which is impossible for a probability. The root with the minus sign is at most $m/m = 1$, by the same inequality, and it is positive because $(m+1)^2 > m^2 - m + 1$. So the estimate is

$$\hat{p} = \frac{m + 1 - \sqrt{m^2 - m + 1}}{m}.$$

Checks

- If $p = 1$ every experiment takes one throw: $\mathrm{E}(N) = \frac{1}{1 \cdot 1} = 1$, and $m = 1$ gives $\hat{p} = (2-1)/1 = 1$.
- The probabilities sum to 1: $\sum_r \mathrm{P}(N = r)$ telescopes to $\lim_{r \to \infty}(1 - q^r)^2 = 1$.
- For a fair die, $p = \frac{1}{6}$ gives $\mathrm{E}(N) = \frac{8/3}{11/36} = \frac{96}{11} \approx 8.73$, a bit more than the 6 throws one die needs on average, as it should be. A simulation of 200,000 experiments gave a mean of 8.71, and putting $m = 96/11$ into the formula for $\hat{p}$ returns $\frac{1}{6}$.

Common slips

- Counting each die's throw separately: a simultaneous throw of both dice counts as *one* throw, which is exactly why $N = \max(X, Y)$.
- Forgetting that the term $pq^{2r-2}(1+q)$ needs the sum with $x = q^2$, not $x = q$.
- Keeping both roots in part (ii) without checking which one can be a probability.

Beyond STEP

The trick $\mathrm{P}(\max \leq r) = \prod \mathrm{P}(X_i \leq r)$ is the standard way to handle maxima (and, with $\geq$, minima) of independent variables, the start of *order statistics*. Part (ii) is the *method of moments*: match the sample mean to the theoretical mean and solve for the parameter. First-year statistics compares it with maximum likelihood and asks how accurate such estimators are. For the probability side, Grimmett and Welsh, *Probability: An Introduction*, is a good next step.

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